

# Peptimancer Vault Report

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## Base Molecule Analysis

Sequence: HAEGTFTSDVSSYLEG

Length: 16 amino acids

Generation ID: dfa0102b-670b-4fd6-91aa-5a37d920f5f0

## Analogue 1: Metha-Series Atylidae

Vault ID: PMNC-A51F8C5

### Modified Sequence:

HAEGTFTSDVSSYLEG -> HAEGTF-DMeth-SDVSSYLEG

### Modifications Applied:

- Substitution of Threonine (T) at position 7 with D-Methionine (DMeth)

### IP Risk Profile:

Patent Risk: Low

Novelty Score: 80.0%

Notes: While Methionine substitutions are variously found in peptide patents, our unique position and selected D-isomer promise patent differentiation.

### Bioactivity Profile:

Binding Affinity: -8.0 kcal/mol

Predicted Half-Life: 3.0 days

Synthesis Complexity: 3/5

Notes: The D-Methionine substitution could preserve DPP-4 resistance while slightly enhancing albumin binding. Synthetic challenges primarily revolve around incorporating the D-amino acid component.

*This report contains proprietary peptide designs. Handle confidentially.*

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